Deepak Bhat

Software Engineer

+91-9964489913 | deepakravindrabhat@gmail.com | linkedin.com/in/deepakrbhat
github.com/deepakbhat18 | deepakbhat.is-cool.dev

Summary

Computer Science and Engineering undergraduate graduating in 2027, with hands-on experience building backend APIs and full-stack applications using Java, Spring Boot, React.js, SQL, and Python. Skilled in REST API development, JWT authentication, role-based access control, database optimization, and machine learning integration. Seeking Software Engineer or Software Developer roles.

Education

Bachelor of Engineering in Computer Science and Engineering **2023 – 2027 (Expected)**
CMR Institute of Technology, Bengaluru *CGPA: 8.72 / 10*

Technical Skills

Programming Languages: Java, SQL, C, JavaScript (Basic), Python (Basic)
Backend Development: Spring Boot, Spring Framework, Spring Security, REST APIs, JWT Authentication, FastAPI, JPA/Hibernate, Resilience4j
Databases: MySQL, PostgreSQL, Database Design, SQL Query Optimization, Indexing
Frontend Development: React.js, HTML, CSS
Machine Learning: Scikit-learn, Flower (Federated Learning)
Tools & DevOps: Git, Maven, Docker, Linux/Ubuntu, Postman, Swagger/OpenAPI, Nginx
Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design Fundamentals, Machine Learning Fundamentals

Projects

OralCare – Dental Clinic Management Platform
Java, Spring Boot, React, PostgreSQL, REST APIs, JPA/Hibernate

- Developed a dental clinic management platform featuring appointment scheduling, real-time queue management, treatment tracking, and patient reporting workflows.
- Built 10+ REST APIs using Spring Boot, Spring Security, JWT authentication, and role-based access control (RBAC) for secure access.
- Improved database access through PostgreSQL indexing and JPA/Hibernate query optimization, identifying and resolving common N+1 query issues.
- Integrated administrative dashboards, automated email notifications, and PDF report generation to streamline routine clinical management workflows.

BookMySlot – Event and Slot Booking Platform **Live Demo**
Java, Spring Boot, React.js, MySQL, REST APIs, JWT

- Developed a full-stack event and slot-booking platform supporting four roles: Admin, Club Staff, Teachers, and Students.
- Implemented 15+ REST APIs for event management, slot booking, approval workflows, and role-based access management.
- Applied MySQL indexing and database normalization to organize booking data and improve query efficiency across application workflows.
- Integrated asynchronous email notifications and analytics dashboards to support event management, booking administration, and activity tracking.

FedMed-AI – Federated Learning Medical Risk Prediction System
Java, Spring Boot, Python, FastAPI, PostgreSQL, Docker, Resilience4j

- Developed a medical risk prediction system covering five disease categories using Scikit-learn machine learning models and structured datasets.
- Implemented federated learning using Flower and FedAvg across simulated hospital datasets for collaborative model training.
- Integrated Spring Boot and FastAPI services for disease prediction, feature preprocessing, and asynchronous model-training workflows.
- Implemented training-job status tracking and optimized PostgreSQL data access using JPA EntityGraphs for efficient entity loading.

Leadership & Achievements

- **1st Place, CSE Department Project Competition (2025):** Won against 50+ department entries with OralCare, a backend system handling real-time data for 200+ daily transactions, using optimized database queries and JWT-secured APIs.
- **Team Lead, Smart India Hackathon 2025:** Led a 6-member team to design a distributed system for waterborne disease tracking, with cloud-ready backend services; selected for technical excellence.
- **Runner-Up, DSA Hackathon, CMRIT:** Placed 2nd among 100+ competitors, solving data structures and algorithms problems under time constraints.